

Notch Filter

1) Objectives of Notch Filter

- Remove narrow-band powerline interference (50 Hz or 60 Hz)
- Attenuate target frequency noise while leaving nearby frequencies unaffected
- Improve signal-to-noise ratio (SNR) for downstream EEG analysis and classification
- Preserve overall EEG signal integrity and neural activity patterns

2) How It works

A Notch Filter is a **band-stop filter** applied to EEG data. It:

- Targets a very narrow frequency band (typically 50 Hz or 60 Hz)
- Attenuates signals at that specific frequency
- Leaves frequencies just above and below mostly unchanged

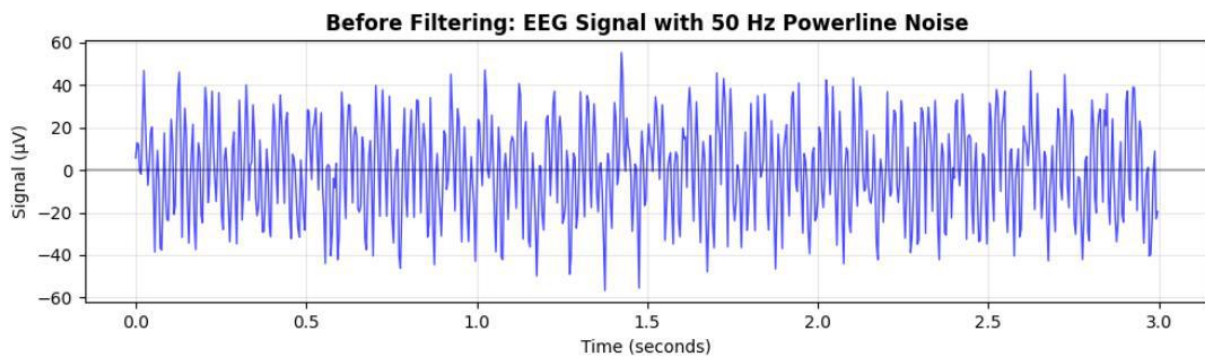


Figure 1 — Power spectral density before Notch Filter. The 50 Hz peak is significantly attenuated.

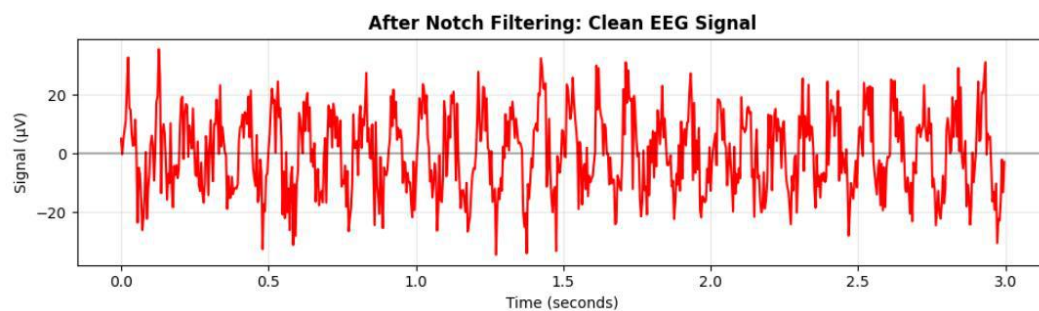


Figure 2 — Power spectral density after Notch Filter. The 50 Hz peak is significantly attenuated.

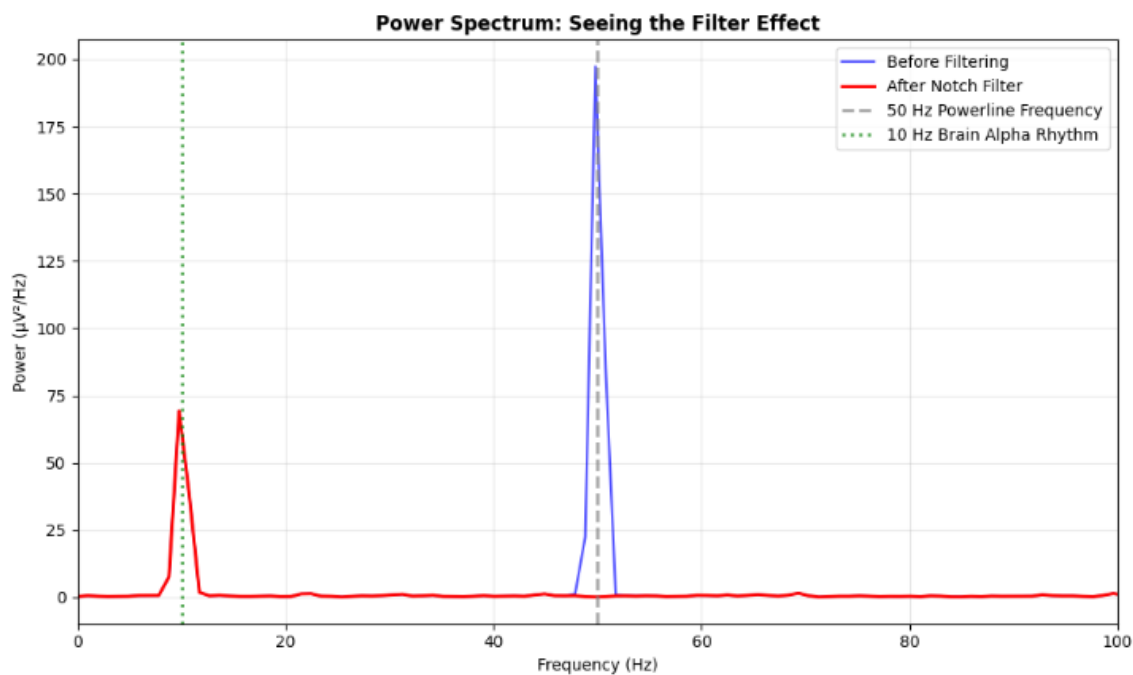


Figure 3 — Raw EEG trace (channel 1) before and after filtering. Powerline oscillation is removed while neural rhythms remain.

